Factored Form of a Quadratic Function

Name _____

Algebra 2 Lesson 2-3 (fast track) Date _____ Period _____

Learning goal: I can find the zeros of a quadratic function by factoring, use them to answer a real-world question, and use them to find where the function is positive or negative.

Language goal: I can explain what the zeros of a function tell me using the words *factor*, *zero*, *Zero Product Property*, *x-intercept*, and *interval*.

On the quiz: You will (1) factor $3x^2 - 11x - 4$ (full Savvas quiz only); (2) solve $x^2 + 7x = 30$ by factoring, the steps of Examples 4 and 5; (3) find when a launched object lands (Example 4, Try It 4; bonus); (4) find where a quadratic function is positive (Example 5, Try It 5); (5) write a long-jumper's parabola (full Savvas quiz only).

On the Topic 2 test: Question 4 asks for a rectangle's width from its area (Example 4; the Exit question is the same kind). Question 7 asks when a ball thrown from a stadium hits the ground (Example 4). Question 14 asks when a launched ball is higher than 12 m, an inequality (Example 5; you also need the Quadratic Formula from Lesson 2-6).

Zero Product Property: if $ab = 0$, then $a = 0$ or $b = 0$ (or both). To use it, rewrite the equation so one side is 0, factor, then set each **factor** equal to 0. Spanish: *Propiedad del cero del producto*; zero of a function = *cero de una función*.

Example 4: Find the Zeros of a Quadratic Function

A multilevel driving range has three levels. Marco hits golf balls from the second level, which is 32 ft high. The height of a ball at x seconds after Marco hits it is modeled by the function $h(x) = -16x^2 + 16x + 32$. When does the ball hit the ground?

1. The ball hits the ground when the height, $h(x)$, is 0:
   $0 = -16x^2 + 16x + 32$
2. Factor out the GCF, -16:
   $0 = -16(x^2 - x - 2)$
3. Factor. Be careful of the signs: two numbers with product -2 and sum -1 are 1 and -2.
   $0 = -16(x + 1)(x - 2)$
4. Zero Product Property (-16 is never 0):
   $x + 1 = 0$ or $x - 2 = 0$, so $x = -1$ or $x = 2$

The **zeros** of the function are at $x = -1$ and $x = 2$. They are the solutions of $0 = -16x^2 + 16x + 32$ and the **x-intercepts** of the graph of h.

Which zero makes sense? Time has to be positive, so $x = 2$ is the only solution that makes sense. After **2 seconds**, the golf ball hits the ground.

Check: $h(2) = -16(4) + 16(2) + 32$
$= -64 + 32 + 32 = 0$ ✓

Say it: After I *factor*, the *zeros* of h are _____ and _____, by the *Zero Product Property*. They are the *x-intercepts* of the graph. The ball hits the ground after _____ seconds, not _____ seconds, because . . .

Try It 4: When does the ball reach the ground?

A baseball is thrown from the upper deck of a stadium, 128 ft above the ground. The function $h(x) = -16x^2 + 32x + 128$ gives the height of the ball x seconds after it is thrown. How long will it take the ball to reach the ground?

Set the height equal to 0: $0 =$ _____

Factor out the GCF: $0 = -16\,($_____$)$

Factor: $0 = -16\,($_____$)($_____$)$

Zeros: $x =$ _____ or $x =$ _____

Which zero makes sense? Why?

_____

_____

_____

The ball reaches the ground after _____ seconds.

Extension (DOK 3): Drone flight (Practice & Problem Solving 41)

Use Structure A drone leaves its launch pad, flies down over the edge of a cliff, and then it flies back up again. The ordered pairs are the drone's flight time and height: (0 s, 5 m), (5 s, 0 m), (10 s, 0 m), (15 s, 5 m).

a. What does each point represent for the drone and its height?

_____

_____

b. Write a function that represents the drone's height with respect to time. Factor your function if it is factorable.

Hint: the *zeros* are 5 and 10, so $h(x) = a(x - 5)(x - 10)$. Substitute the point $(0, 5)$ to find a.

_____ _____

_____ _____

$h(x) =$ _____

c. How far below the platform does the drone fly?

_____ _____

_____ _____

_____ _____

d. How could you validate this model?

_____ _____

_____ _____

_____ _____

Example 5: Determine Positive or Negative Intervals

Identify the interval(s) on which the function $y = x^2 - 2x - 3$ is positive.

The y-values of a quadratic function can only turn from positive to negative or from negative to positive when the graph crosses the x-axis. Find the **zeros** to find these points.

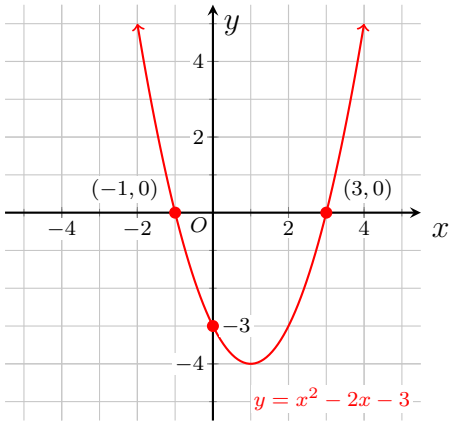
$0 = x^2 - 2x - 3$

$0 = (x - 3)(x + 1)$

$x - 3 = 0$ or $x + 1 = 0$

$x = 3$ or $x = -1$

Two zeros create three **intervals**. Choose an x-value to test in each interval.



$x < -1$	$-1 < x < 3$	$x > 3$
Choose $x = -3$.	Choose $x = 1$.	Choose $x = 6$.
$y = (-3)^2 - 2(-3) - 3$	$y = (1)^2 - 2(1) - 3$	$y = (6)^2 - 2(6) - 3$
$= 9 + 6 - 3 = 12$	$= 1 - 2 - 3 = -4$	$= 36 - 12 - 3 = 21$
Positive	**Negative**	**Positive**

The function is positive when the graph is above the x-axis, on the intervals $(-\infty, -1)$ and $(3, \infty)$.

Say it: I *factor* to find the *zeros* _____ and _____. They are the *x-intercepts*, and they cut the x-axis into three *intervals*. The function is positive on _____ because . . .

Try It 5: Where is the function negative?

Identify the interval(s) on which the function $y = x^2 - 4x - 21$ is negative.

Factor: $0 = (_____)(_____)$ Zeros: $x =$ _____ and $x =$ _____

Interval	_____	_____	_____
Test x			
y			
Sign			

The function is negative on the interval _____.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
TRU	Solve a quadratic equation using the zero product property	Warm-up for Example 4 (optional): the factors are already given
NMZ ★	Factor out a monomial	Quiz 3 (bonus): the -16 step of Example 4 and Try It 4
CJC ★	Solve a quadratic equation by factoring	Quiz 2 and Quiz 3: Example 4 and Try It 4
DP9 ★	Graph solutions to quadratic inequalities	Quiz 4: Example 5 and Try It 5
UB5	Factor quadratics	Quiz 1 (full Savvas quiz only): factor $ax^2 + bx + c$

Optional, not on the short form: G2Q Write a quadratic function from its zeros and HPB Write a quadratic function in factored form from its x-intercepts and another point (Quiz 5, full Savvas quiz only; also the drone extension).

Before the quiz, finish these sentences.

To find when a launched object hits the ground, I set $h(x)$ equal to _____, *factor*, and use the *Zero Product Property*. If $0 = -16(x + 2)(x - 5)$, then $x =$ _____ or $x =$ _____; in a time problem I keep only _____.

Each *zero* of a quadratic function is an *x-intercept* of its graph because

Two zeros cut the x-axis into _____ *intervals*. To tell whether the function is positive on an interval, I

Exit question. The length of a rectangle is 5 in. longer than its width and the rectangle has an area of 50 in.^{2}. What are the dimensions of the rectangle? Without graphing, how do you know which *zero* is a possible solution?

